

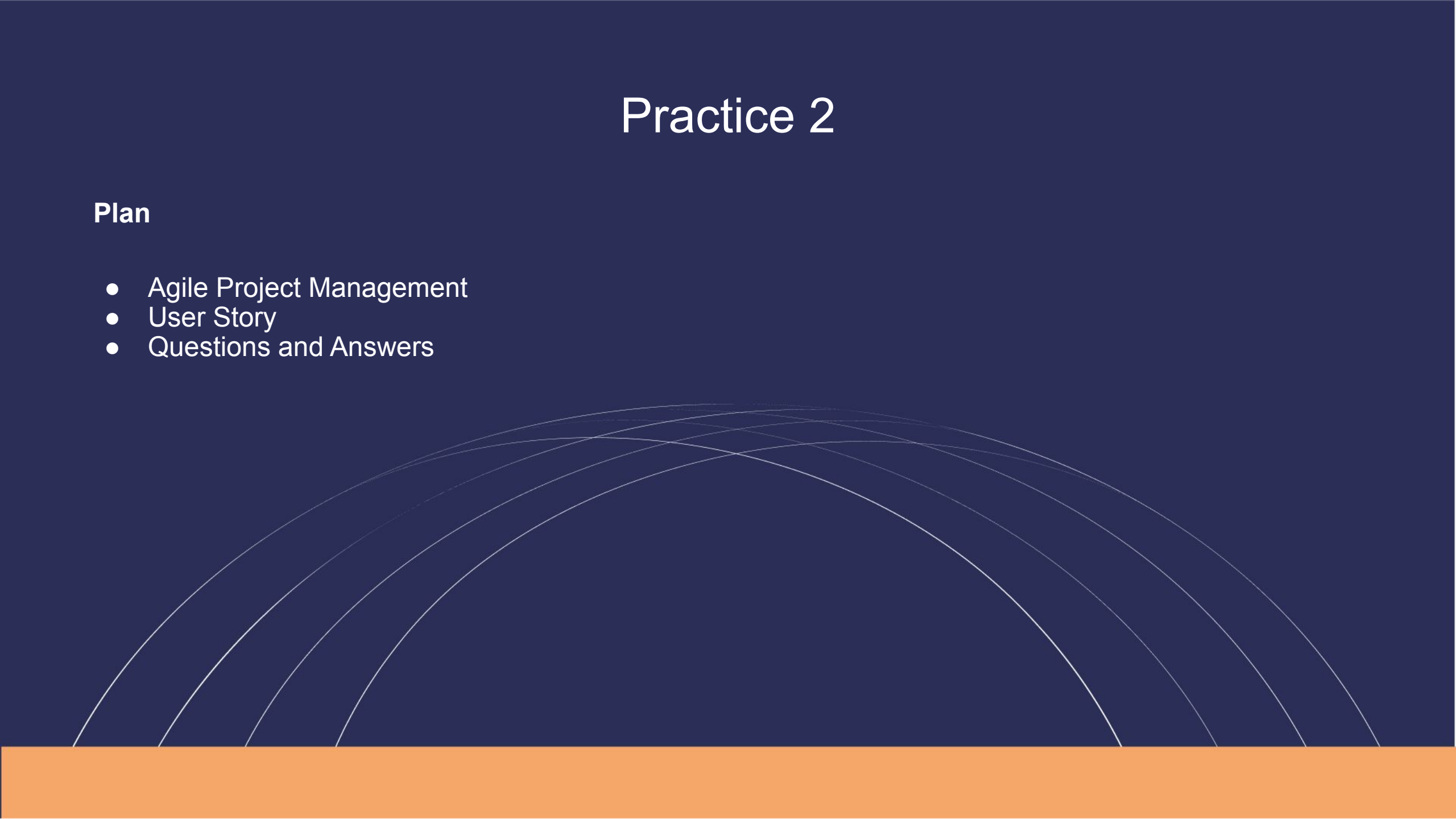


INF 451 - Project Management IS

Fall 2026

Practice 2

Plan

- Agile Project Management
 - User Story
 - Questions and Answers
- 

- [Practice lesson](#)
- [Grading sheet](#)
- [Google form](#)

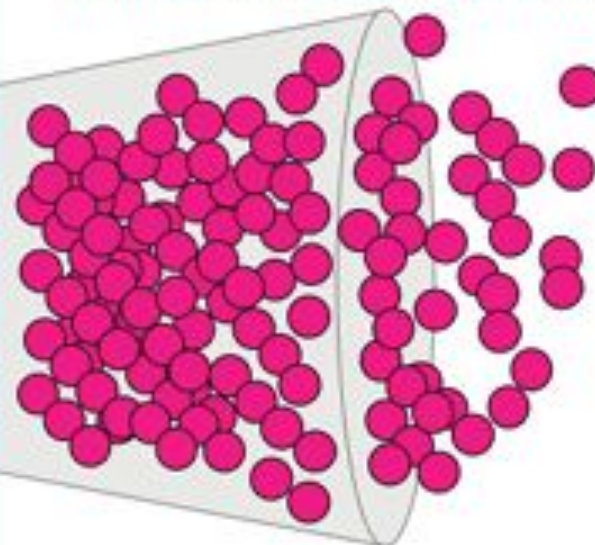


AGILE IS A
MINDSET

DESCRIBED BY
4 VALUES

DEFINED BY
12 PRINCIPLES

MANIFESTED THROUGH
UNLIMITED NUMBER OF PRACTICES



KANBAN

SCRUM

SAFe®



MINDSET

VALUES

PRINCIPLES

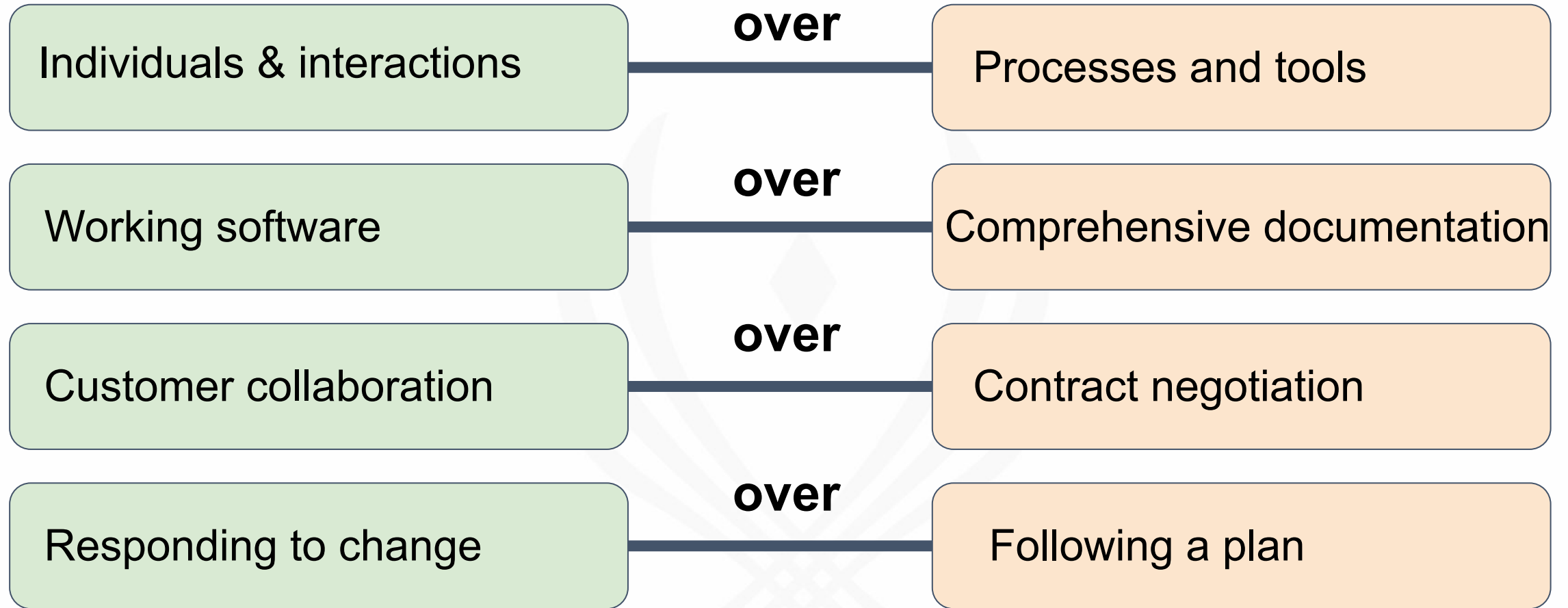
PRACTICES

Agile Project Management

Agile is a method for delivering value, most often through a Project.



- Through this work we have come to value:



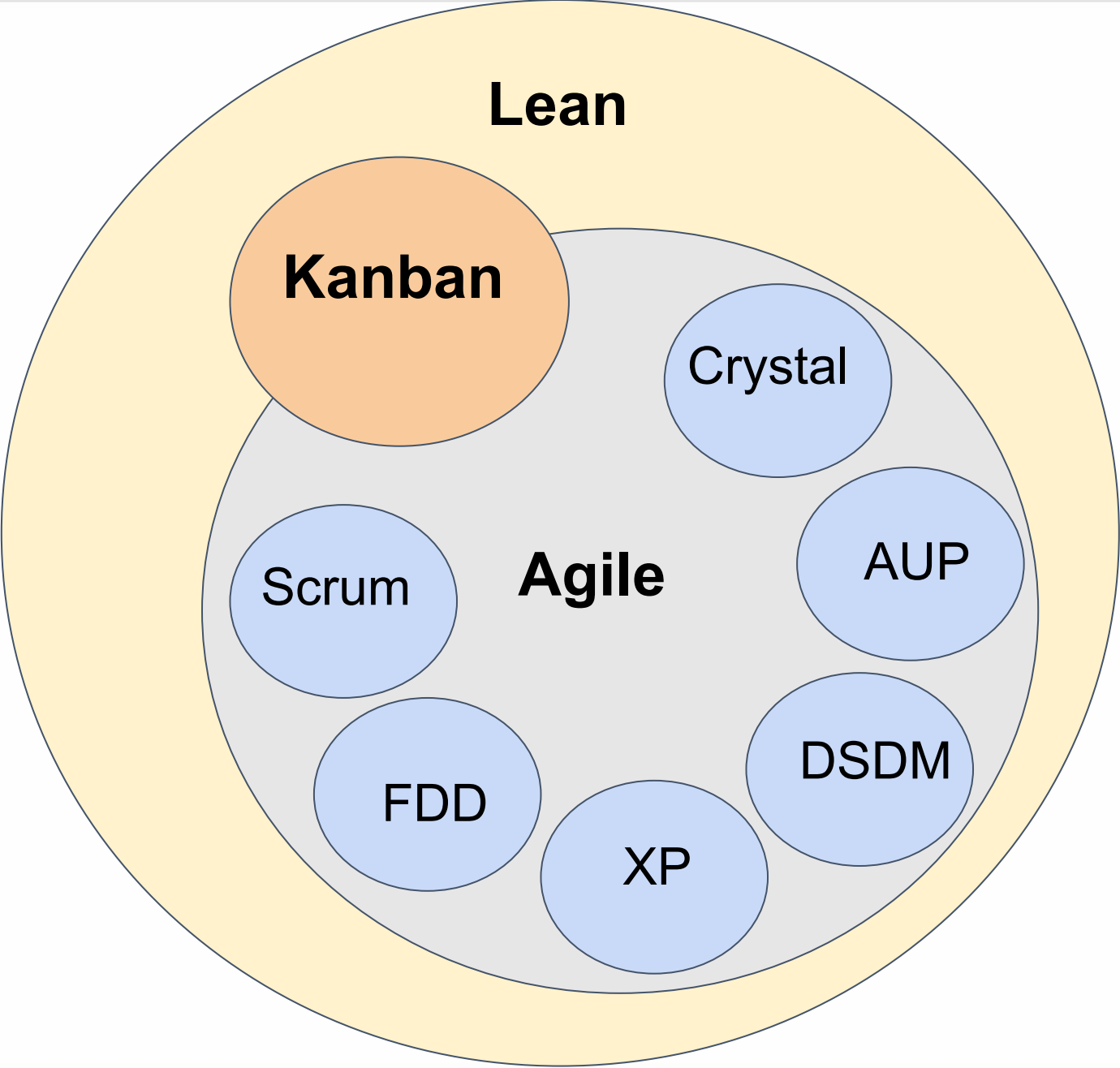
*That is, while there is value in the items on the right, we value the items on the left more.
These are supported by **the 12 clarifying principles**.

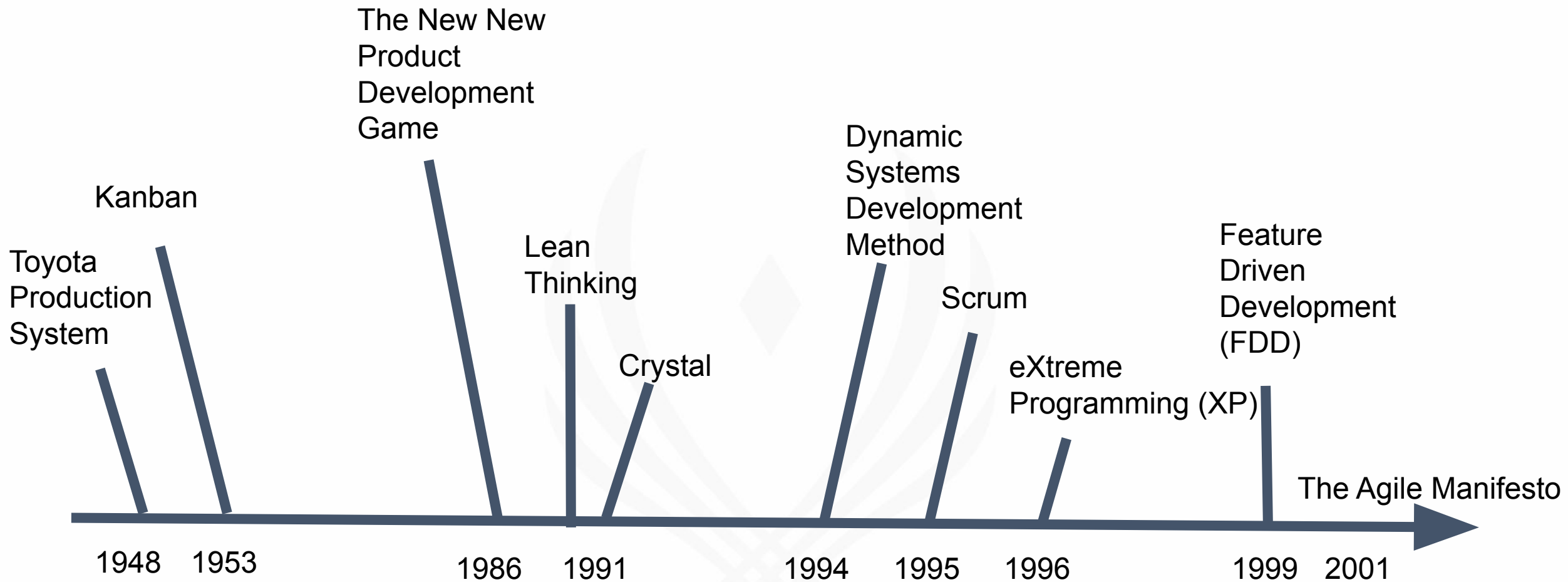
Agile Principles and Mindset

The Agile Manifesto

- In 2001, a group of 17 individuals representing the most widely used lightweight software development methodologies, agreed on a common set of values and principles which becomes known as the **Agile Manifesto**.
- Uncovering better ways of developing software by doing it and helping others do it.

History of Agile

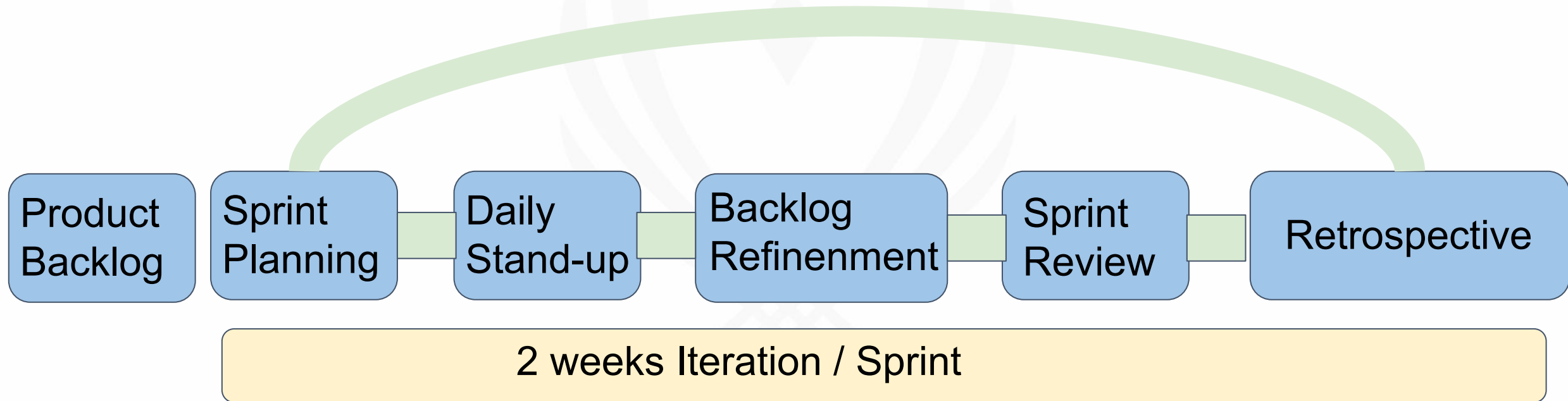




*Agile has gained in popularity as knowledge work and high- uncertainty work has increased over 20 years.

Agile Overview

Typical Agile approach:



Agile Roles

- **Product Owner**

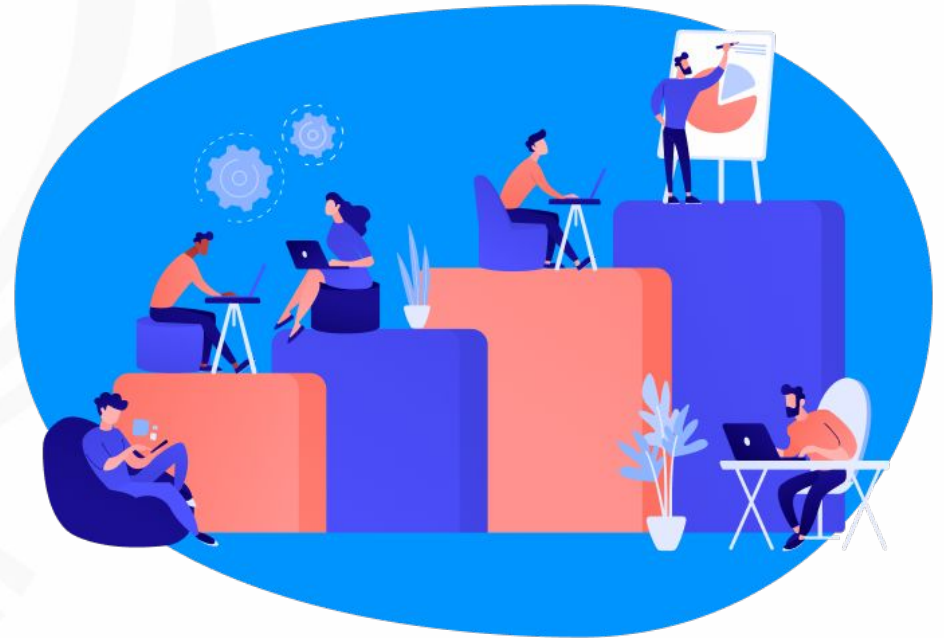
- Represents the customer,
- Provides or approves new features in the **Product Backlog**,
- Prioritizes the features in the Product Backlog.

- **Scrum Master**

- The Agile “Servant Leader”
- Facilitates the ceremonies,
- Is a neutral third party to facilitate problem solving.

- **Team**

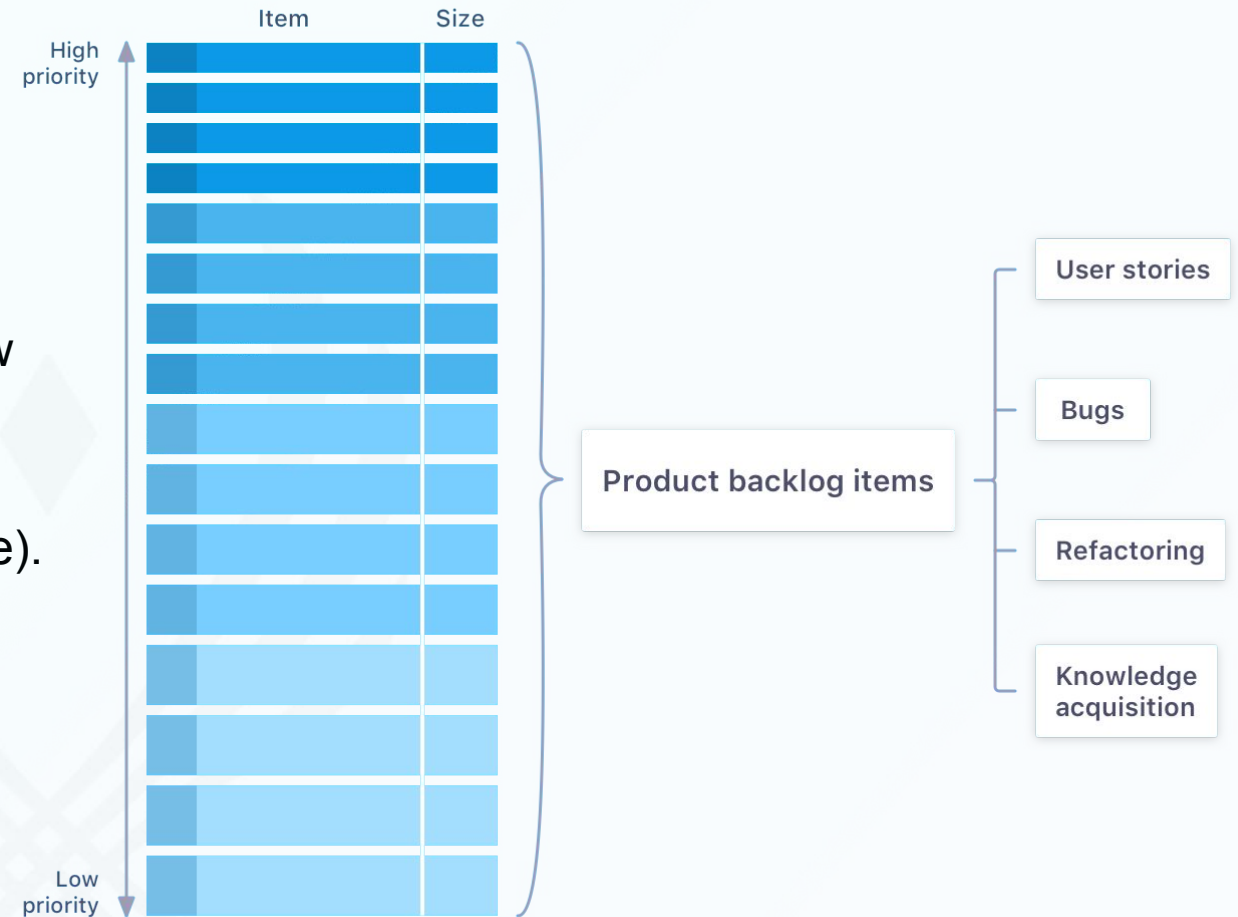
- Often Developers.
- Perform the work & create value.



Product Backlog

- **Product Backlog**

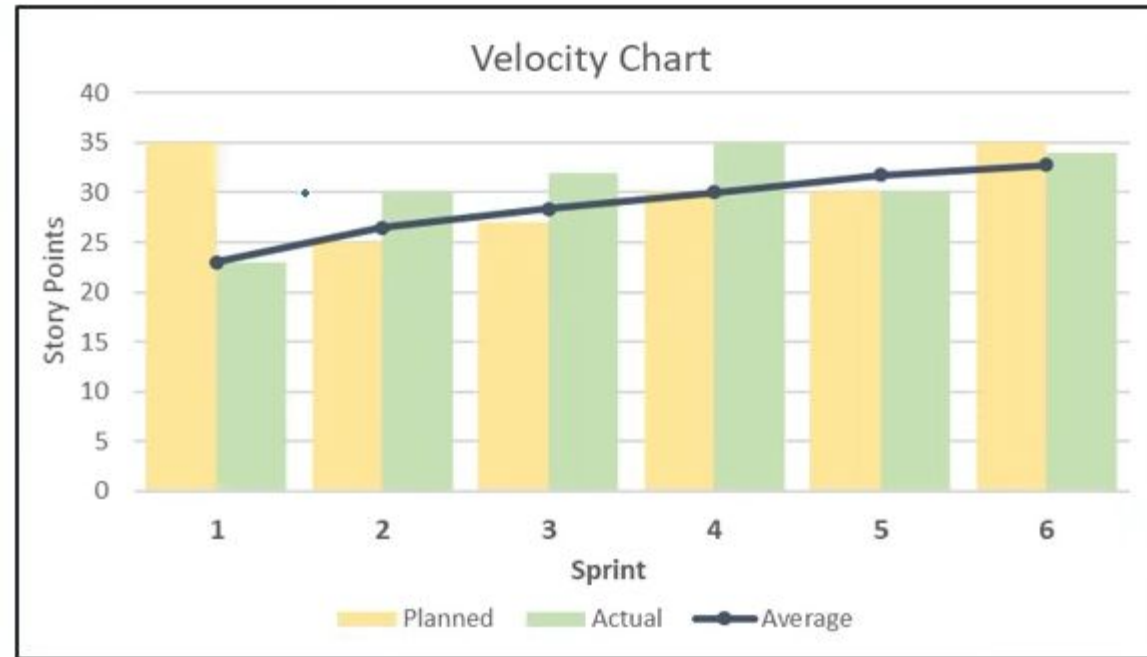
- A list of customer-valued **features**,
- Each can be delivered in a few weeks to a few months,
- Created or approved by the **Product Owner**,
- **Prioritized** (often from most to least valuable).



- The next feature is broken down into smaller pieces called **User Story Cards** by the team.
- These have the Acceptance Criteria - what needs to be completed - and are established for effort in **Story Points**.

- **Sprint Planning**

- The team selects the next highest priority **User Stories**, enough to be completed in a sprint.
- The number is matched to how much the team can complete in a sprint - their **Velocity**.



- **Daily Stand-up**

- The team gathers around their Kanban Board, and gives an update on progress, or any blockers for their stories.
- Around 15 min.
- Show progress.



- **Backlog Refinement**

- The “Three amigos” - Customer (Senior User or PO), Developers, and Testers - meet when necessary to:
- break down more features into User Stories,
- give them Acceptance criteria, and:
- estimate their effort.

We need enough to be ready for the next sprint.



- **Sprint Review**

- A usable increment or feature should be ready -
The team demonstrates this to the customer, who gives feedback and/or approves.



- **Retrospective**

- The team meet to reflect on their process, and what can improve.
- They ask:
- What went wrong?
- What challenges did we face?
- What did we learn?
- What still puzzles us?



The 12 Claryfing Principles



1) Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.



2) Welcome changing requirements, even late in development. Agile processes harness change for the the customer's competitive advantage.



3) Deliver working software frequently, from a couple of months, with a preference of the shorter timescale.



4) Business people and developers must work together daily throughout the project.



5) Build project around motivated individuals. Give them the environment and support they need, and trust them to get the job done.



6) The most efficient and effective method of conveying information to and within a development team is face to face conversation.



7) Working software our primary measure of progress.



8) Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.



9) Continuous attention to technical excellence and good design enhances quality.



10) Simplicity - the art of maximizing the amount of work not done - is essential.



11) The best architectures, requirements, and designs emerge from self organizing teams.



12) At regular intervals, the team reflects on how to become more effective, then tunes and adjust its behaviour accordingly.

Scrum

Scrum is a process framework used to manage product development and other knowledge work.

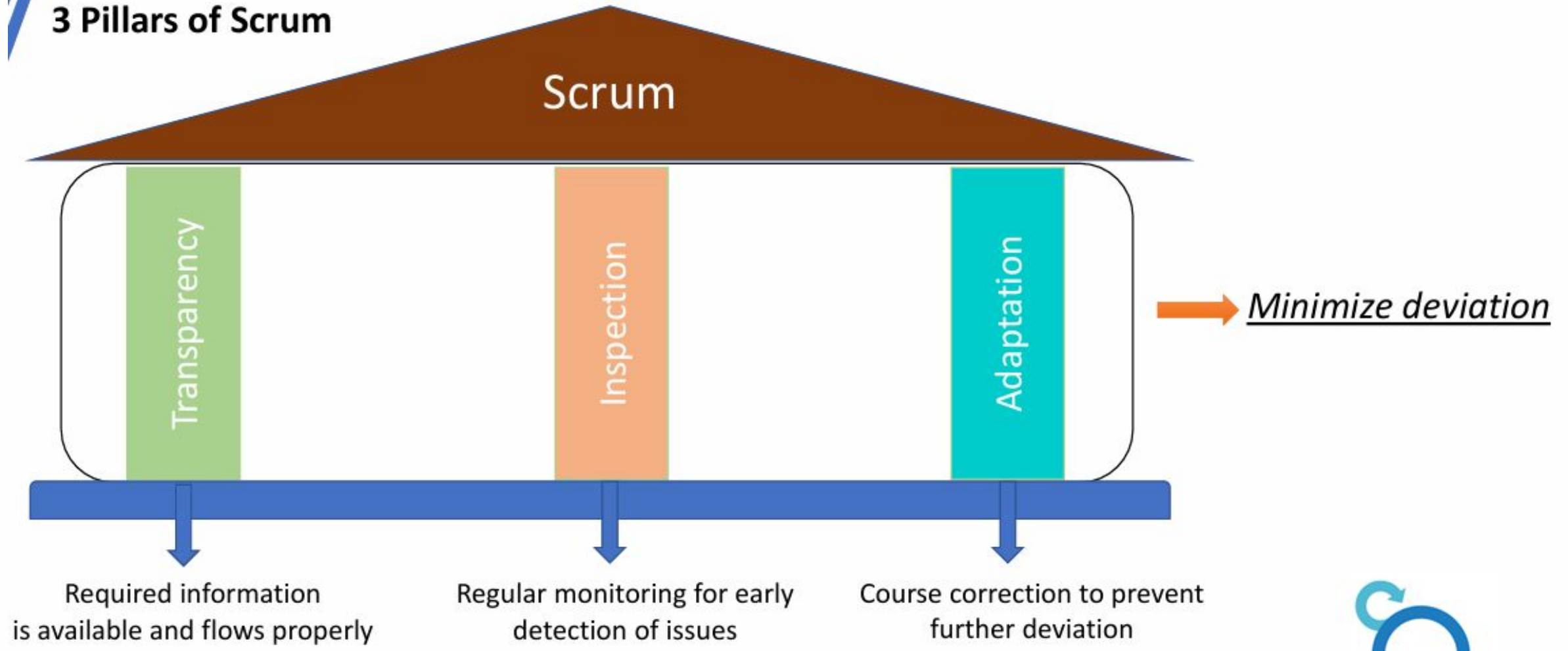
- *Agile Alliance*

- Scrum is a set of rules for structuring the team, processes and techniques for making the development process agile..
- Scrum is a lightweight framework designed to help people, teams, and organizations generate value by finding adaptive solutions to complex problems.

Foundation of the Scrum



3 Pillars of Scrum





Sprint

Sprint Planning



Sprint
Retrospective

Daily Scrum



Sprint Events

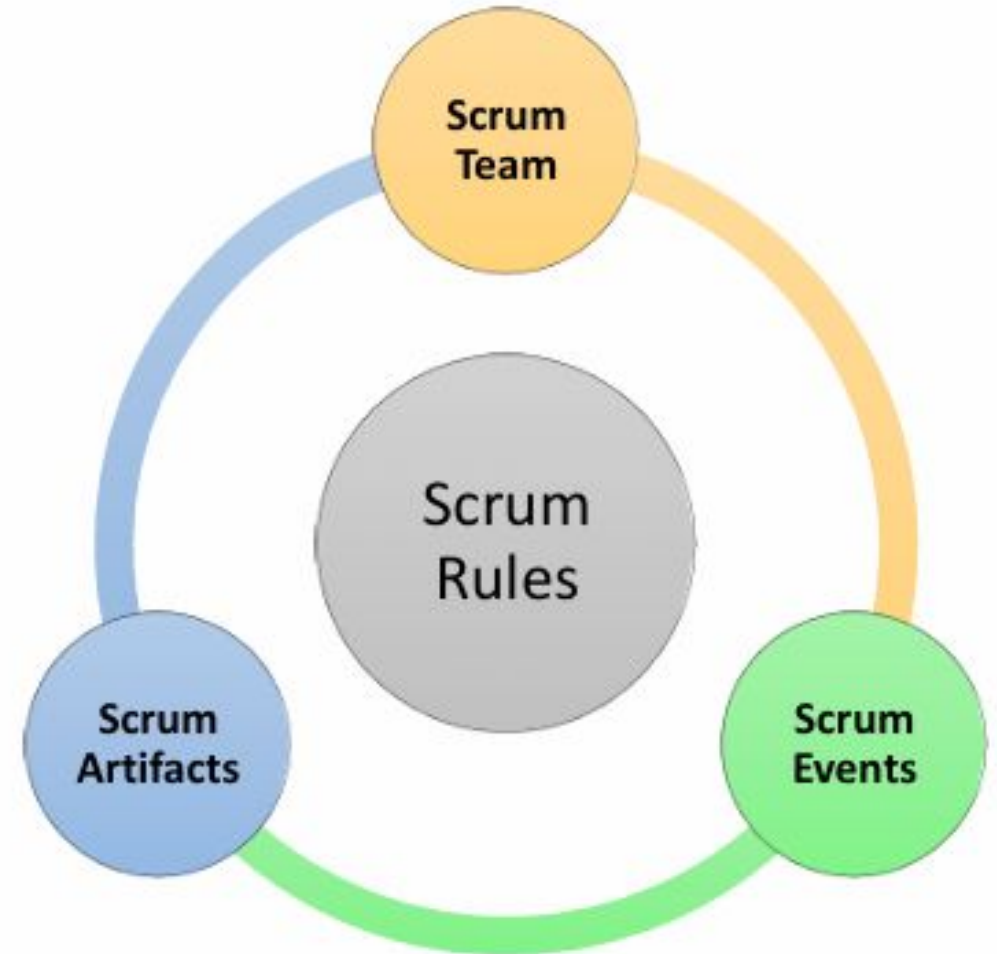
Sprint Review



Scrum elements

- 1. Scrum artifacts: Product Backlog, Sprint Backlog, Increment => Scope (What Scrum teams need to create?)
- 2. Scrum team: Product Owner, Development Team, Scrum Master => (Who is behind the scrum magic?)
- 3. Scrum events (How to advance the project toward to the desired outcome?)

4 components of Scrum Framework



User story

User Story

Acceptance Criteria

- Should be testable
- Should be clear and concise
- Everyone must understand
- Should provide user perspective

User Story Examples

- As a [customer], I want [shopping cart feature] so that [I can easily purchase items online].
- As a [manager], I want to [generate a report] so that [I can understand which departments need more resources].
- As a [customer], I want to [receive an SMS when the item is arrived] so that [I can go pick it up right away].
- As a [manager], I want to be able to [understand my colleagues progress], so I can [better report our success and failures].



Example of a good User Story

As an [online visitor], I want to [add products in my shopping cart] so that [I can purchase multiple products at one go]

Acceptance Criteria [Abstract]

- Products can be added to the cart
- Products can be removed from the cart.
- Shopping cart will be empty initially
- Shopping cart will be empty after purchase
- Products can be added with multiple quantities in the cart
- Shopping cart will show the total product breakdown quantity and cost with grand total



Epic

- Series of user stories with a broader strategic objective
- A large user story that cannot be delivered as defined within a single iteration or is large enough that it can be split into smaller user tasks/user stories
- No standard format to represent epics



User story template

- <https://docs.google.com/spreadsheets/d/1Be7j8iUMkOoLqofL102YvYBFXpK7zNkh/edit?usp=sharing&ouid=107431048622011300376&rtpof=true&sd=true>

#story — unique story number.

story_title — short name of the user story.

story description — what the user wants to accomplish and why.

story_estimation / range out of __ — estimated effort, e.g., 5/10 or 8/13.

HC&HV — High Complexity & High Value.

LC&HV — Low Complexity & High Value.

HC&LV — High Complexity & Low Value.

LC&LV — Low Complexity & Low Value.

User_role — e.g., Student, Teacher, Administrator, Customer.

Pre-Requirements — conditions that must be completed before starting the story.

QA Test — how the story will be tested/verified.

Responsible person in a team — team member responsible for implementation.

Iteration — Sprint/Iteration number.

Deadline — planned completion date.

Materials

- Week 1 Materials:
- <https://drive.google.com/drive/folders/17isFKpsD6FCw18yh9Gy3O00PRB2-SfPu>
- Course Materials:
- https://drive.google.com/drive/folders/16K14MloBo7IErvVuj0ao7IM2xSVVcOf_

Submit

- Create your Trello or Notion or Kanban board in order to follow your progress in one place
- Sprint 1 planning
- Submit to moodle
- Product Backlog with User Stories (Template)

Questions

